NICHOLAS TRIGGER

Biomedical Engineer

(361) 446-3005 | nicktrig124@gmail.com | linkedin.com/in/nick-trigger | nicholastrigger.com

SUMMARY

Biomedical Engineering alumnus from Duke University specializing in medical device R&D, with hands-on experience designing, prototyping, and validating hardware from PCB-level electronics to mechanical enclosures. Comfortable across embedded systems (C/C++, RTOS, ESP32) and benchtop testing under ISO/FDA-conscious development, with clinical research and leadership experience that bridges engineering and real-world clinical workflows.

EDUCATION

Duke University, Durham, NC — **AUG 2022 – MAY 2026**

Batchelor's of Science in Biomedical Engineering

Relevant Coursework: Biomaterials, Biomechanics, Medical Device Design, Embedded Medical Devices, Imaging Systems

EXPERIENCE

Research and Development Engineer — **DEC 2025 – MAY 2026**

Reselute Medical (Durham, NC)

- Designed and validated test protocols to support product verification activities and design history file documentation, in alignment with FDA and ISO 13485 requirements.
- Conducted literature reviews and risk based analysis to inform and refine experimental designs and test protocols.
- Documented and analyzed experimental results, facilitating informed design decisions and technical reporting supporting design history file documentation.

Classroom Research and Development Engineer — **APR 2025 – MAY 2026**

Duke University Biomedical Engineering Dpt. (Durham, NC)

- Designed, fabricated, and assembled electronic laboratory hardware for deployment across BME teaching labs.
- Developed a stimulus-responsive EKG simulator generating physiologically accurate signals for lab use.
- Validated hardware performance under extended use conditions to ensure durability and safety.

Clinical Research Associate — **JUN 2023 – AUG 2024**

Christus Spohn Health System (Corpus Christi, TX)

- Supported emergency medicine clinical studies by ensuring compliant data collection and thorough EMR review.
- Collected and documented patient consent in accordance with IRB and FDA guidelines.
- Collaborated effectively with physicians, residents, and research staff to facilitate ongoing clinical investigations.

PROJECTS

Central Line Associated Bloodstream Infection Prevention Device — *nicholastrigger.com/projects/clabsi*

- Developed handheld UV-C disinfection device for central-line hubs as Duke BME senior capstone, targeting 40,000 annual CLABSIs and eliminating dependency on nurse compliance with Scrub-the-Hub protocol.
- Validated device performance through design verification testing, achieving a 4-log CFU reduction and 2.49 mW optical output targeting the 263-273 nm germicidal window.
- Engineered three custom PCBs, an SLA & FDM 3D printed enclosure, and designed firmware to operate the device safely.

Radial Arterial Line Placement Training Device — *nicholastrigger.com/projects/arm/*

- Co-developed a durable, anatomically accurate arterial line placement training device to address the shortage of realistic, reusable simulation tools for radial a-line insertion training.
- Iterated through multiple prototype revisions incorporating clinician/user feedback to improve pulse simulation fidelity, tissue realism, and durability under repeated needle insertions.

CERTIFICATIONS AND SKILLS

Certifications: CSSC: *Lean Six Sigma White Belt*, Nordic Semi: *nRF Connect SDK Fundamentals*, *Matter Fundamentals*

Technical Skills: Design Controls, Design History File (DHF), ISO 13485, ISO 14971, Risk Management, FMEA, BioSignal Processing, X-ray, MRI, CT, PET, Medical Device Prototyping, Biomechanics, ISO Standards, FDA Regulations, EMG, Medical Image Processing, Medical Firmware, GD&T, PCB Layout, PCB Design, Python, C, C++, Embedded C, FreeRTOS, Zephyr, JavaScript, TypeScript, HTML/CSS, SQL, SPI, I2C, UART, CAN, BLE, USB, V&V, Bench Testing, System Validation, Bacterial Testing, Phantoms, Data Collection & Analysis, PCB Triage, Prototyping

Tools: SolidWorks, Fusion 360, AutoCAD, OnShape, SolidWorks, KiCAD, EAGLE CAD, Git, GitHub, GitLab, 3D Printing, Laser Cutting, CNC Milling, Manual Machining, Waterjet Cutting, PCB Assembly, PCB Fabrication, Microsoft Office, LaTeX